Daily Paper Review: Pass the Baton — Trajectory-Relayed On-Policy Distillation

Internbot Literature Review

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1 Paper Summary

1.1 Problem and Motivation

Standard on-policy distillation (OPD) trains a student by having it generate complete reasoning trajectories on-policy, then supervising with teacher log-probabilities [1]. The fundamental problem is **prefix failure**: when a student commits to an incorrect reasoning direction early in generation, all subsequent tokens are conditioned on a flawed premise that the teacher would never have produced. The teacher's supervision signal on these downstream tokens becomes low-quality noise, degrading the training signal.

Prior interventions fall short: FastOPD uses fixed-length truncation regardless of *where* reasoning actually fails; TRD (Trajectory-Refined Distillation) rewrites trajectories offline after rollout, but repairs contain visible artifacts and actually degrade performance to 30.69% vs. OPD's 41.23%; SKD (Speculative Knowledge Distillation) performs token-level mixing but cannot detect reasoning-level failures or break repetitive loops. Preliminary experiments show that replacing only the *single reflection token* at each failure point lifts accuracy from 27.73% to 34.96% (+7.23%), demonstrating that corrections can be remarkably local and that early intervention is far more valuable than late intervention.

1.2 Method

Handoff trigger criterion. Relay-OPD detects reasoning failure points via a **label-free** trigger based on teacher-student asymmetry on reflection tokens:

$$\phi(h) = \mathbf{1}[a^T(h) \in \mathcal{R}] \cdot \mathbf{1}[\mathcal{K}_S(h) \cap \mathcal{R} = \emptyset] \tag{1}$$

where $a^T(h) = \arg\max_v \pi_T(v|h)$ is the teacher's top-1 token, $\mathcal{K}_S(h) = \mathrm{TopK}_K(\pi_{\bar{\theta}}(\cdot|h))$ is the student's top-K support ($K = 5$), and $\mathcal{R}$ is a curated set of 13 reflection tokens ("Wait," "But," "Hmm," "Actually," etc.). The trigger fires when the teacher recognizes the need to redirect reasoning while the student does not—requiring no ground-truth answers, reward models, or external verifiers.

Relay trajectory construction. The trajectory interleaves student-generated segments ("student legs") with brief teacher-generated corrections ("teacher legs") via a relay race metaphor. A state machine governs three states $s_t \in \{S, T, \mathrm{term}\}$:

- **Student leg** ($s_t = S$): student generates tokens autoregressively; at each position, $\phi(h)$ is evaluated.
- **Handoff** ($\phi = 1$, budget remaining): teacher "takes the baton," generating L paragraphs of corrective reasoning starting from its reflection token.

- **Resumption**: after L paragraphs, the student resumes generation from the corrected state.
- **Budget**: at most M teacher takeovers; after the M-th, the trajectory terminates.

The optimal relay budget is $(M = 2, L = 3)$—at most 2 teacher interventions of 3 paragraphs each.

Training objective. The student is supervised via a PPO-style clipped objective on the relay trajectory tokens z_t (from both student and teacher legs):

$$\mathcal{L}_{\text{Relay}} = -\mathbb{E}_{x,z}\left[\frac{1}{N}\sum_{t=1}^{N}\min\Big(\rho_t \cdot A_t^{\text{Relay}},\ \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\cdot A_t^{\text{Relay}}\Big)\right] \tag{2}$$

where $\rho_t = \pi_\theta(z_t|h_t^z)/\pi_{\bar{\theta}}(z_t|h_t^z)$ is the importance ratio and $A_t^{\text{Relay}} = \log \pi_T(z_t|h_t^z) - \log \pi_{\bar{\theta}}(z_t|h_t^z)$ is the advantage. On teacher-leg tokens, the advantage is strongly positive (teacher assigns high probability to its own corrections while student assigns lower), providing a clear learning signal.

Unified speculative decoding engine. Rather than running separate generation pipelines, Relay-OPD implements relay generation as state-switched speculative decoding: the student serves as the draft model, and during teacher legs, drafts are verified against π_T with acceptance probability $\min(1, \pi_T/\pi_{\text{student}})$. Teacher logits computed during verification provide the trigger criterion $\phi(h)$ at *zero extra cost*.

Self-diminishing intervention. As the student improves during training, fewer triggers fire, naturally reducing teacher involvement from $\sim$13% of tokens initially to 2–3% after 20 steps—an emergent transition toward standard on-policy generation.

1.3 Results

Experiments use Qwen3-4B teacher with Qwen3-1.7B and Qwen3-0.6B students on 8 mathematical reasoning benchmarks (AIME 2024/2025/2026, MATH500, AMC23, OlympiadBench, HMMT Feb/Nov), trained on DAPO-Math-17K with 8×H100 GPUs.

Main results. Qwen3-1.7B student: Relay-OPD achieves **46.96%** average vs. OPD at 41.23% (+5.73 pp), FastOPD at 45.47% (+1.49 pp), SKD at 42.35% (+4.61 pp), and TRD at 30.69%. Qwen3-0.6B student: Relay-OPD achieves **31.04%** vs. OPD at 28.03% (+3.01 pp) and FastOPD at 30.42% (+0.62 pp). SKD *degrades* the 0.6B student to 24.38% (below OPD).

Ablations. (1) Teacher leg value: adding 3-paragraph teacher corrections atop trigger-based truncation provides +2.77 pp. (2) Relay-token objective outperforms teacher FKL (+2.88 pp) and student-draft-token (+2.40 pp) variants. (3) Top-K for trigger: $K = 5$ is optimal; $K = |\mathcal{V}|$ recovers standard OPD (41.23%), confirming the trigger is essential. (4) Budget: $M = 2$ optimal; $M = 4$ degrades by -2.95 pp (too off-policy). (5) $L = 3{-}4$ optimal; $L = 5$ degrades.

Efficiency. Training trajectory length reduced by **50.7%** (1.7B) and **63.9%** (0.6B) vs. OPD. Convergence at step 35 vs. step 55 for OPD. Relay-OPD produces both shorter and more accurate responses (e.g., AIME26: -14.2% length, +4.17% accuracy vs. FastOPD).

1.4 Limitations

The authors acknowledge: (1) evaluation limited to mathematical reasoning with Qwen3 models; (2) the reflection token set $\mathcal{R}$ may require adjustment for different model families or reasoning styles; (3) effectiveness depends on a sufficient teacher-student capability gap—benefits diminish as the gap narrows; (4) the relay budget (M, L) was tuned for the 1.7B student and reused for 0.6B without re-tuning.

1.5 Relevance to Our Research

This paper directly extends our extensive **on-policy distillation** (Topic 1) review series. Our prior reviews have covered single-turn OPD variants (Direct-OPD, Filter-Then-Reweight, DOPD, Early Stopping Rollout, ReNIO), multi-turn extensions (ReOPD), and cross-architecture distillation (TIDE, AR-to-Diffusion OPD). Relay-OPD introduces a qualitatively new intervention point: *online, reasoning-state-driven* trajectory construction during generation, rather than post-hoc filtering or fixed truncation. The label-free trigger criterion based on reflection-token asymmetry is a practical contribution that avoids the reward model dependency of many RL-based approaches. The self-diminishing teacher ratio (13% $\rightarrow$ 2–3%) provides an elegant automatic curriculum, connecting to our review of ReOPD's reliability-aware scheduling and the Physics paper's three-region framework.

2 Follow-up Research Directions

2.1 Direction 1: Extending Relay-OPD to Multi-Turn Agentic Distillation

Gap. Relay-OPD is designed for single-turn mathematical reasoning where the entire trajectory is a monolithic chain-of-thought. In multi-turn agentic settings (code execution, search, tool use), the student's failure modes are qualitatively different: wrong tool selection, incorrect parameter binding, or misinterpretation of environment observations. The reflection-token trigger set $\mathcal{R}$ ("Wait," "But," etc.) is specific to reasoning redirection and would not detect agentic failures.

Approach. Design domain-adaptive trigger criteria for multi-turn relay distillation: (1) for tool-use tasks, define $\mathcal{R}_{\text{tool}}$ as the set of tool-call correction patterns (e.g., "Let me try a different function," "That returned an error") and trigger when the teacher's top-1 token initiates a tool correction while the student's top-K does not; (2) for code execution, trigger on teacher-student divergence at the first token *after* receiving an execution result (where the student must decide whether to debug or proceed); (3) combine with ReOPD's prefix replay [2]: use relay trajectories as the prefix pool, where teacher legs serve as pre-recorded corrections that ReOPD can replay without environment interaction. This hybrid eliminates both the environment cost (via prefix replay) and the prefix failure problem (via relay corrections).

Feasibility. High. The trigger mechanism generalizes naturally—only the token set $\mathcal{R}$ and the trigger context change, not the relay construction or training objective. ReOPD's existing framework for pooling teacher trajectories directly accommodates relay trajectories. A pilot on the math+tool environment from ReOPD's experiments would test whether relay prefixes outperform standard teacher prefixes for prefix replay.

2.2 Direction 2: Learned Trigger via Subliminal Clock Signals in Diffusion LLMs

Gap. Relay-OPD's trigger relies on a hand-crafted reflection token set $\mathcal{R}$ that is model-family-specific and may not transfer across architectures. For diffusion LLMs (which generate via iterative denoising rather than left-to-right), reflection tokens have no natural analog—there is no sequential generation to "redirect." Yet diffusion LLMs still suffer from quality degradation during denoising, and a relay-like correction mechanism could improve their generation quality.

Approach. Replace the reflection-token trigger with a learned trigger based on the subliminal clock signal [3]: (1) during MDLM denoising, extract the internal τ representation (denoising progress) from the model's mean activation vectors; (2) define a "denoising failure" trigger as a mismatch between the model's internal τ estimate and the expected τ for the current step—when the model "thinks" it is further along than it actually is, it may be overcommitting to low-quality token choices; (3) at triggered positions, inject teacher corrections by re-masking the affected tokens and letting a stronger teacher MDLM re-denoise them. This creates a "diffusion relay" where the student denoises most positions but the teacher corrects critical ones, analogous to Relay-OPD's student/teacher leg alternation.

Feasibility. Moderate. The subliminal clock provides a continuous, model-intrinsic signal that generalizes across diffusion LLMs without domain-specific token sets. The main challenge is defining the mismatch threshold for triggering correction. The paper's finding that >90% of τ variance concentrates in a single PC provides a clean 1D signal for threshold-based triggering. A prototype using LLaDA as student and SDAR-8B as teacher on text generation benchmarks would test whether diffusion relay improves over standard MDLM generation.

2.3 Direction 3: Relay-Guided Skill Induction for Self-Play

Gap. Skill Self-Play [4] evolves skills through an exploration stream that discovers novel task patterns, but the skill controller can only induce skills from *successful* exploration outcomes. When the solver fails on a task, the failure is discarded—no skill is induced from the failure mode. Relay-OPD's trigger mechanism identifies exactly *where* and *how* reasoning fails, but this diagnostic information is used only for trajectory correction, not for systematic skill acquisition.

Approach. Feed Relay-OPD's trigger diagnostics into Skill Self-Play's skill induction pipeline: (1) during solver training on Skill-SP's curriculum, run Relay-OPD to identify trigger points in the solver's trajectories; (2) cluster trigger points by the teacher's corrective pattern (what type of reasoning redirection was needed—e.g., constraint re-evaluation, alternative decomposition, numerical verification); (3) for each cluster, the skill controller induces a "defensive skill" that encodes the failure pattern, the correction strategy, and validators checking for the specific error type; (4) the proposer uses these defensive skills to generate targeted training tasks that exercise the solver's weak points. This closes the loop between failure detection (Relay-OPD) and curriculum construction (Skill-SP), making the self-play system actively *learn from its own failures* rather than discarding them.

Feasibility. High. Both systems are independently validated. The trigger clustering requires only collecting (prefix, trigger-point, teacher-correction) triples during relay generation and running the existing skill controller on these triples. The defensive skill format follows Skill-SP's standard 6-tuple structure. A pilot on the mathematical reasoning domain (shared by both papers) would test whether relay-induced defensive skills improve solver performance on problems that previously triggered corrections.

3 Conclusion

Relay-OPD introduces a principled mechanism for constructing hybrid training trajectories that interleave student-generated segments with brief, surgically targeted teacher corrections at reasoning failure points. The label-free handoff trigger—based on teacher-student asymmetry on reflection tokens—is simple, requires no external supervision, and adds zero computational overhead through the unified speculative decoding engine. The method achieves +5.73 pp over standard OPD while reducing training trajectory length by 50–64% and converging 36% faster. The self-diminishing teacher ratio provides an elegant emergent curriculum, with teacher involvement naturally declining from 13% to 2–3% as the student improves. For our distillation research program, Relay-OPD fills a specific gap between fixed truncation (FastOPD) and post-hoc rewriting (TRD): *online, state-driven intervention* during generation. The three proposed follow-up directions—multi-turn agentic relay via ReOPD integration, diffusion relay via subliminal clock triggers, and relay-guided skill induction for self-play—each connect Relay-OPD's failure detection mechanism with complementary systems from our review portfolio.

References

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